Federated Multi-Agent Reinforcement Learning for Heterogeneous Action Spaces

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***Abstract*—The utility of multiple reinforcement learning (RL) agents collaboratively training within a shared environment, all working towards common objectives, is increasingly evident within the Internet of Vehicles (IoV). The multi-agent Advantage Actor-Critic (MA2C) algorithm is a prominent example of such a Multi-Agent Reinforcement Learning (MARL) system. However, MA2C requires agents to share policies, such as pairs of states and actions and even trained models, among neighboring agents, to overcome the challenge of agents having only partial observations. Unfortunately, this requirement amplifies the communication overhead and raises privacy concerns. Federated learning (FL), as a privacy-preserving machine learning method, can be applied in the MARL context with a central server aggregating the weights of the agents' models. However, this technique assumes that all agents are capable of executing identical actions, which may be impractical. In this paper, we introduce a novel FL A2C algorithm called Advantage Actor Federated Critic (A2FC). The proposed algorithm streamlines the aggregation of agents' critic models while offloading the training of actor models to the individual agents' local machines. An experiment conducted in an adaptive traffic signal control (ATSC) system demonstrates the method's effectiveness in personalizing agents' actions, preserving agents' privacy during training, and mitigating communication overhead issues.**

I. Introduction

RL is a systematic mathematical framework for self-guided learning driven by experiences, defined within the context of a Markov Decision Process (MDP). RL trains autonomous agents to learn optimal behaviors in interactive environments. Deep reinforcement learning (DRL), first introduced in [1], extends RL's capabilities by integrating deep learning models and RL techniques. RL can be categorized into value-based methods, policy-based methods, and actor-critic methods. *Q learning* [2], a prime example of a value-based method, uses a state-action value function that is updated incrementally based on experiences. However, such methods are "off-policy", updating policies using one-step *temporal differences*, which assumes stationary MDP transitions. In contrast, policy-based methods like *REINFORCE* [3] are "on-policy", updating policies using sampled episodic returns. Actor-critic methods [4] combine aspects of both value-based and policy-based methods. They separate the policy function from the value function, thus reducing bias and variance compared to policy-based approaches. An improvement is Advantage Actor-Critic or A2C [5], which introduces an estimated advantage value to evaluate how good an action is at the current state.

With the proliferation of IoT, IoV and smart devices, there is a growing demand for multiple RL agents to collaboratively train in shared environments to achieve global objectives. These environments can be characterized as cooperative MARL systems. Despite the potential benefits, existing MARL algorithms face certain challenges. Traditional MARL algorithms, based on Q-learning, distribute the global Q-function to local agents [6], [7]. These approaches rely on strict rules to balance scalability and optimality. In contrast, independent Q-learning [8] enhances scalability as each local agent independently trains its policy, treating others as parts of the observable environment. However, the scalability of Q-learning is constrained by its tabular structures, especially in real-world scenarios involving resource-constrained devices. Moreover, tabular structures struggle to leverage previous experiences or shared knowledge in similar states.

The A2C algorithm is considered an advanced approach compared to Q-learning in a large-scale MARL system. Agents apply the A2C algorithm within the multi-agent framework to achieve faster convergence [9]. The introduction of FL [10]–[12], a concept that has gained considerable attention in the field of privacy-preserving machine learning, has expanded the potential applications of algorithms like A2C, which rely on neural networks. FL involves the coordination of multiple clients, such as computers and smart sensors, by a central server to collectively train a machine learning model. There have been efforts to combine FL with DRL to enhance its robustness to similar but previously unseen states [13], [14]. However, these approaches raise concerns about the required communication overhead and privacy leakage.

An example MARL deployment scenario is an adaptive traffic signal control (ATSC) system where multiple RL agents work together to dynamically adjust traffic signal timings, aiming to reduce traffic congestion across the entire area. The area is composed of multiple administrative regions, each containing several intersections. Each RL agent is a resource-constrained device capable of sensing and detecting traffic flow and average speed, and is responsible for controlling only one signalized intersection. The traditional MARL methods mentioned above are unsuitable for various reasons in this ATSC

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system. Q-learning-based methods are resource intensive and hence unsuited for deployment on resource-constrained devices. Furthermore, every agent needs to share its observations and policies with other agents, and receive others' as inputs, or send and receive both actor and critic models in the federated learning settings. This results in a significant communication overhead when agents must make rapid real-time decisions. Additionally, each intersection may have different traffic rules, for example, some intersections allow left or right turns, while others may only permit straight-ahead traffic, and traditional FL aggregation struggles to address the issue of agents having a heterogeneous action space. Finally, privacy leakage can also be a concern if the ATSC system is subject to rules and regulations limiting information sharing.

To address the above challenges, we propose a new federated A2C framework called Advantage Actor Federated Critic (A2FC). Unlike a traditional application of FL in the MARL system where a central server averages the weights of the agents' actor and critic models separately and sends the average weights back to the agents, the proposed framework only aggregates the weights of the agents' critic models. Our proposed framework is especially suited to MARL systems where agents have heterogeneous action spaces, such as the ATSC system. Also, the proposed framework does not require any communication among agents, which significantly reduces the communication overhead, and protects the privacy of agents' policies in MARL systems.

The main contributions of this paper can be summarized as follows: we propose a novel federated A2C algorithm for MARL systems with agents having heterogeneous action spaces. The proposed method averages the agents' critic neural network models only. It has the benefits of 1) allowing agents to have heterogeneous action spaces and personalizing agents' optimal actions in their local environments; 2) reducing the communication overhead as agents in the MARL system do not need to communicate with each other; 3) preserving privacy as agents do not share their observations or policies with others. We demonstrate the effectiveness of our proposed method in a comprehensive experiment in an ATSC system.

II. Preliminaries And Challenges

A. Policy-based RL Algorithms

The Policy Gradient method, also known as REINFORCE, enhances a parameterized model π_θ by employing gradient descent to maximize the cumulative reward over the long term. The parameters θ are iteratively adjusted to increase the likelihood of selecting a sequence of optimal actions, utilizing sampled total rewards. Assuming a trajectory $(s_1, a_1, r_1, s_2, a_2, r_2, \cdots, r_T)$ on the parameterized model π_θ, the loss function is formulated as follows:

$$\mathcal{L}(\theta) = \hat{R}_T \sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) \tag{1}$$

Here, the estimated total reward is denoted as $\hat{R}_T = \sum_{t=1}^{T} \gamma^t r_t$.

The A2C algorithm integrates the characteristics of Q-learning into the features of the policy gradient method. It introduces a value function V_ω to estimate the expected reward $\mathbb{E}_\pi[R_t^\pi|s_t = s]$. The estimated total reward is revised to $\tilde{R}_T = \hat{R}_T + \gamma^T V_\omega(s_T)$. The algorithm defines the Advantage function as $A_t = \tilde{R}_t - V_\omega(s_t)$, which is utilized to update the neural network parameters of both the Actor (θ) and the Critic (ω). The Advantage function effectively assesses the performance of the agent's action in the current step by comparing it to other possible actions.

B. Multi-agent Reinforcement Learning

MARL focuses on the behavior of multiple agents within a shared environment, where all agents collectively observe the environment's state and receive rewards based on their joint actions. Consider a network $G(\mathcal{V}, \mathcal{E})$ that represents interactions among multiple agents in a MARL system. Each agent $i \in \mathcal{V}$ can take discrete actions $a_i \in \mathcal{A}_i$, communicate with node j if edge $(i, j) \in \mathcal{E}$, and access a global reward $r(s, a)$. Many MARL algorithms are grounded in the context of Q learning. These algorithms allow agents to observe the global state and collaboratively take actions that contribute to a decomposable global Q function $Q(s, a) = \sum_{i\in\mathcal{V}} Q_i(s_i, a_i)$. However, merely combining MARL with Q learning disregards agent interactions, resulting in sub-optimal global convergence performance. Coordinated Q learning strikes a balance between optimality and scalability through iterative message passing and control synchronization among neighboring agents. In this approach, $Q_i(s, a) \approx \sum_{j\in\mathcal{N}_i} M_j(s, a_j, a_{\mathcal{N}_j})$, where $\mathcal{N}_i$ refers to the set of neighbors of agent i, and M_j represents the message from neighbor j of agent i. Independent Q learning is a fully scalable strategy but without message passing. In this case, all local Q functions are dependent solely on the local action, i.e. $Q_i(s, a) = Q_i(s, a_i)$. Each agent must know other agents' policies and treat all other agents' behavior as part of the environment dynamics in continuous policy training.

Q learning inherently relies on tabular structures with limited numbers of states and actions due to real-world device storage and computation constraints. Independent A2C extends the principles of independent Q learning to the actor-critic framework where agents still share the global state and reward. Each agent individually trains its policy π_{θ_i} and the corresponding value function V_{ω_i}, estimating the local return as $R_{t,i} = \hat{R}_t + \gamma^t V_{\omega_i}(s_t|\pi_{\theta_i})$. Federated A2C further extends Independent A2C from the multi-agent environment to the FL scenario using the FedAvg algorithm. In this approach, each agent updates its policy θ_i and value function ω_i parameters locally, which are then aggregated on a centralized server. The server computes the mean parameters, i.e., $\theta = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} \theta_i$ and $\omega = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} \omega_i$, and sends them back to the agents for training in the next round.

C. Challenges

While MARL algorithms have advanced significantly over the past decade and have found widespread practical applications, there are still important limitations. One of the key issues

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is that existing MARL algorithms often overcome partial observations at each agent through extensive communication among neighboring agents. This can lead to slow convergence and hence practically prevent the system achieving global optimality through joint actions of the agents. Moreover, the communication among agents results in high latency and communication overhead, rendering the algorithms unsuitable for many applications. Additionally, the assumption that agents can freely communicate and share states and trained policies does not always hold in practice due to either computational limitations or regulatory restrictions. Agents may therefore only be able to train their policies independently based on their own observations, making the assumption of full communication unrealistic. Finally, policy sharing raises privacy concerns, particularly in scenarios where sensitive data is involved. This makes it challenging to implement MARL in problems where data privacy is a priority.

The FL framework offers a solution through the sharing of client model parameters, but not client data, for aggregation on a central server but the current federated A2C approach necessitates the transmission of both policy and value function parameters. This introduces significant communication overhead and latency, which could be impractical in applications where real-time decision-making is crucial. In summary, while MARL algorithms have made strides, these challenges of partial observation, slow convergence, communication overhead, privacy concerns, and latency still remain.

III. Advantage Actor Federated Critic (A2FC)

A. Overview of Method

To address the aforementioned challenges, we propose a novel framework called Advantage Actor Federated Critic (A2FC). In this framework, agents are required to upload only the parameters of their value functions (critic models) to a central server for aggregation, rather than transmitting both policy model (actor model) and value function parameters. The schematic overview of our proposed framework is illustrated in Figure 1.

Consider a scenario with M agents working collaboratively within an environment. At each step, every agent observes the current state of the environment and selects an action that advances the system to the subsequent state. The environment subsequently provides each agent with a reward based on its chosen action, which is then used to update both the agent's critic and actor models. The critic model approximates the expected reward using the obtained reward and subsequently guides the actor model towards making optimal decisions. Once agents have iteratively updated their models for a predetermined number of iterations, they transmit their value function parameters to a central server designated for aggregation. At the central server, the parameters of the agents' critic models are averaged, and the resulting average parameters are sent back to all agents and then used to initialize the next training iteration. This iterative process continues, thereby enabling the collaborative refinement of the agents' critic models within the A2FC framework.

Our proposed approach offers several key advantages over previous MARL algorithms. First, unlike traditional federated A2C methods, our approach accommodates agents within MARL systems that possess heterogeneous action spaces. This flexibility allows agents to have distinct action options, which is a departure from the conventional assumption of uniform action spaces. Secondly, our method streamlines communication, necessitating agents to interact solely with the centralized server, as is characteristic of FL techniques. There is no requirement for agents to communicate with one another or possess knowledge about the system's topology or neighborhood. This simplifies model optimization processes and reduces overall complexity. Thirdly, the approach also minimizes communication overhead significantly. Agents are solely tasked with uploading their value function parameters to the server and retrieving the averaged parameters subsequently. This streamlined communication process ensures efficient information exchange, improving system efficiency. Our approach also aligns with the privacy-preserving principles of the FL framework [15]. Agents retain their individual policies and do not communicate these policies with other agents. This design ensures privacy and safeguards sensitive information.

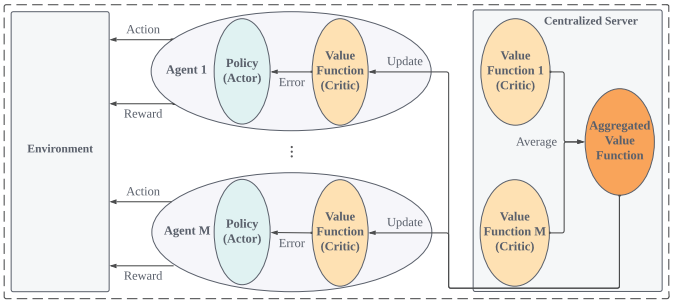


Fig. 1. Overview of A2FC

B. Multi-agent Environment Formulation

Notation and definitions related to MARL were introduced in Section II-B. Here we develop the problem formulation further, with more technical details. Within the network $G(\mathcal{V}, \mathcal{E})$, each agent $i \in \mathcal{V}$ engages in an individual RL task, characterized by an underlying MDP tuple $\langle \mathcal{S}_i, \mathcal{A}_i, \mathcal{T}_i, \mathcal{R}_i \rangle$. Each agent's observation capabilities are confined to its localized environment, encompassing only a portion of the complete environment. The combined coverage of all agents will comprise the entire system's environment by definition. Throughout each time step, all agents collaboratively explore their respective environments and concurrently make decisions. Notably, an individual agent's choices can impact the observable environment and conditions for other agents. During exploration, agent i obtains a reward r_i from the environment, which serves as feedback for the actions it took in each exploration round. This exploration experience is then logged as a tuple within the agent's memory.

In this MARL system, the agents employ A2C algorithms to accomplish their respective RL tasks. Each agent i owns an actor neural network model governed by parameters θ_i

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and a value function neural network governed by parameters ω_i. The role of value functions is twofold: they enhance agents' comprehension of the environment dynamics and reward structure, while also assisting actor models in optimal action selection. Thus, each agent's objective revolves around updating both the actor and value function neural networks with information learned from their exploration experiences. This adaptation process facilitates the acquisition of optimal policies π_{θ_i}, resulting in the selection of actions well-suited to the environmental states encountered in the future.

C. Perspective Of Single Agent

In contrast to previous MARL systems that assume sharing of global rewards and states among agents, we adopt a more flexible approach to address communication overhead and privacy concerns in specific applications. In our approach, agents can access their local rewards derived from the actions they execute. This approach eliminates the need for sharing a global reward, while maintaining a global reward function. Specifically, agents estimate local returns using the following expression:

$$\tilde{R}_{t,i} = \hat{R}_{t,i} + \gamma^t V_{\omega_i}(s_t|\pi_{\theta_i}) \tag{2}$$

Importantly, the gradient of the value function $\nabla\mathcal{L}(\omega_i)$ remains consistent for each agent. This consistency arises from the fact that $\hat{R}_{t,i}$ is sampled using the same stationary policy $\pi_{\theta_i^-}$ [1] for each agent i. The value function $V_{\omega_i} : \mathcal{S} \times \mathcal{A}_i \rightarrow \mathbb{R}$ is responsible for estimating the future return for the policy π_{θ_i}, and this estimation plays a crucial role in computing the gradient of the actor model $\nabla\mathcal{L}(\theta_i)$.

In the proposed MARL system, each agent's local policy and value function are solely dependent on the input state s_t. The loss function for the value function is formulated as follows:

$$\mathcal{L}(\omega_i) = \frac{1}{2T}\sum_{t=1}^{T}(\tilde{R}_{t,i} - V_{\omega_i}(s_{t,i}))^2. \tag{3}$$

Here, T represents the total number of steps, $\tilde{R}_{t,i}$ is the locally estimated return for agent i at time t, and $V_{\omega_i}(s_{t,i})$ stands for the value function's prediction for the state $s_{t,i}$.

The advantage value is defined as $A_{t,i} = R_{t,i} - V_{\omega_i^-}(s_t)$, and based on this definition, the loss function for the actor model is given by:

$$\mathcal{L}(\theta_i) = \frac{1}{2T}\sum_{t=1}^{T}\log\pi_{\theta_i}(a_{t,i}|s_{t,i})\cdot A_{t,i}. \tag{4}$$

In this equation, $\pi_{\theta_i}(a_{t,i}|s_{t,i})$ represents the probability assigned by the actor model to the action $a_{t,i}$ in state $s_{t,i}$. The loss function seeks to maximize the probability-weighted advantage value, which essentially encourages the policy to prefer actions that lead to higher advantages. However, the abilities of agents in this system are limited as each one observes only a portion of the environment. Consequently, they generally cannot collectively optimize global performance without a form of communication, i.e., agents can only achieve local and not global optimization. To overcome this challenge and enable the merging of agents' local observations in a way that respects privacy, we propose using FL.

Traditional Federated MARL methods typically involve aggregating agents' actor models and value functions. However, this approach assumes that all agents have the the same set of actions, denoted as $\mathcal{A}_i = \mathcal{A}j$ for any two agents i and j. This requirement stems from the necessity for the agents' neural network structures to be identical for FL weight aggregation to make sense. In practical scenarios, such as in smart traffic control systems, agents may have differing action spaces due to individual variations such as traffic intersections with differing permissible signals. This diversity of action spaces poses a challenge to traditional FL methods in MARL systems.

D. A2FC algorithm

Algorithm 1 A2FC Algorithm

Require: T, E, η_ω, η_θ
Initialize $s_0 = \{s_{01}, ..., s_{0i}\}$, $B = \{B_1 = \phi, ..., B_i = \phi\}$, $\pi_0 = \{\pi_{01}, ..., \pi_{0i}\}$, $t \leftarrow 0$, $e \leftarrow 0$,
repeat
  $t \leftarrow t + 1$
  for $i \in \mathcal{V}$ **do**
    Select action $a_{t,i} \leftarrow \pi_{t,i}$; Obtained reward $r_{t,i} \leftarrow \mathcal{R}_i(s, a_{t,i})$
    The next state $s'_{t,i} \leftarrow \mathcal{T}_i$; $B_i \leftarrow B_i \cup \{(t, s_{t,i}, \pi_{t,i}, a_{t,i}, r_{t,i}, s'_{t,i})\}$
  end for
  if $t = T$ **then**
    for $i \in \mathcal{V}$ **do**
      Estimate $\tilde{R}_{\tau,i}, \hat{R}_{\tau,i}, \forall\tau \in T$;
      Update ω_i with $\eta_\omega\nabla\mathcal{L}(\omega_i)$; Update θ_i with $\eta_\theta\nabla\mathcal{L}(\theta_i)$
      $s_i \leftarrow s_{0i}, \pi_i \leftarrow \pi_{0i}, t \leftarrow 0$
    end for
    $e \leftarrow e + 1$
  end if
  if $e = E$ **then**
    $\omega = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}}\omega_i$; $\omega_i \leftarrow \omega, \forall i \in \mathcal{V}$
    $e \leftarrow 0$
  end if
until Training Ends

The A2FC algorithm we propose introduces a notable departure from conventional FL methods. Unlike previous approaches, A2FC does not require agents to possess matching action spaces or an equal number of actions. Its innovation lies in the integration of the FedAvg algorithm, taken from FL, into MARL systems where we only aggregate the weights of agents' critic models instead of both actor and critic models. This unique design allows agents to independently train their actor models, enabling personalized action improvement within their distinct exploration domains. Simultaneously, the aggregation of critic models tackles the issue of agents having access only to partial observations. The averaged critic model weights encapsulate all agents' experienced states and their understanding of the environment's reward setting. Algorithm 1 provides a concise overview of our proposed A2FC framework within the context of a MARL system.

E. Method Discussion

The critic model plays a crucial role in helping agents comprehend the reward structure of the environment. By taking an agent's current state or observations as input, the critic model predicts an expected reward achievable through

[1] θ_i^- is a frozen current model used to update θ_i.

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the agent's chosen action. The primary objective of a critic model is to maximize this expected reward within the given state. In a MARL environment, the reward function remains consistent across all agents, even though their exploration is limited to certain parts of the environment. Agents' training experiences can contribute to each other's understanding of the global environment. The FedAvg algorithm enhances agents' awareness in a MARL system by averaging the critic model weights from all agents. This aggregation incorporates agents' past training experiences and their collective comprehension of the reward mechanism from a diverse range of visited states. When the centralized server provides agents with the averaged model, they gain insights into the global environment through others' experiences. This averaged critic model offers improved generalization for agents, especially when encountering unfamiliar states that other agents have previously encountered.

As a result, the loss function in Eq. 3 transforms to:

$$\mathcal{L}(\omega_i) = \frac{1}{2T}\sum_{t=1}^{T}(\tilde{R}_{t,i} - V_\omega(s_{t,i}))^2, \tag{5}$$

where $\omega = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}}\omega_i$. This average critic model enhances agents' ability to cope with new scenarios, thus contributing to more effective learning in a cooperative MARL system.

In contrast, agents' actor models need to retain a more personalized character in order to determine their optimal policies. The actor model operates by taking the agent's current state or observations as inputs and generating a probability distribution across the agent's action space. Conventional FL aggregation of agents' actor models is clearly not desirable when agents possess differing action spaces. In this case, each agent's actor model must be tailored to their individual operating environment, and enforcing a shared action space to fit the FL framework will run counter to that goal, leading to worse performance.

From a privacy preservation perspective, our proposed method ensures that agents' trained policies remain private. Since our approach does not require direct communication among agents, our method enhances the security of agents' actor models by enabling them to be trained individually without any policy sharing. We consider that an agent's actor model contains a more substantial portion of private information compared to their critic model. An actor model updates based on the agent's exploration experiences, encompassing tuples involving visited states, chosen actions, acquired rewards, and ensuing states stored in the agent's memory. Consequently, the actor model directly exposes the agent's learned policy, raising privacy concerns. On the contrary, while a critic model also depends on agents' states or observations as inputs, its focus is centered on estimating the broader environmental reward structure, which is publicly available to all agents. Therefore, the aggregation of critic models results in less privacy leakage from the agents. By design, our approach mitigates privacy risks related to policy exposure, making it a promising solution for secure MARL systems.

IV. Experiments

This section outlines the experimental implementation of the A2FC algorithm within an ATSC environment, utilizing the microscopic traffic simulator SUMO [16]. Our experimental setup draws inspiration from a previous study [9]. For evaluation purposes, we contrast our proposed approach against two cutting-edge methods in the field of ATSC, namely Multi-Agent Advantage Actor-Critic (MA2C) and Independent Advantage Actor-Critic (IA2C). These benchmarks provide a comprehensive comparison to gauge the effectiveness of our A2FC algorithm within the ATSC context.

A. Traffic Grid and Parameter Setups

ATSC systems involve multiple intersections acting as RL agents to collaboratively manage traffic networks, with the goal of reducing potential congestion by adjusting signal timings based on real-time traffic dynamics. In our experiment, the traffic network is designed as a 5x5 synthetic traffic grid, illustrated in Fig. 2, with a total of 25 intersections forming the set of agents. In this traffic grid, two-lane arterial streets have a speed limit of 20 m/s, while one-lane avenues have a speed limit of 11 m/s. There are five possible phases: East-West straight, East-West left-turn, and three phases for straights and left-turns in the East, West, and North-South directions. The action space for each intersection includes either in part or in full these five phases, determined by geographical or traffic rule constraints Additionally, the grid incorporates four time-varying traffic flow groups, accommodating varying levels of traffic demands.

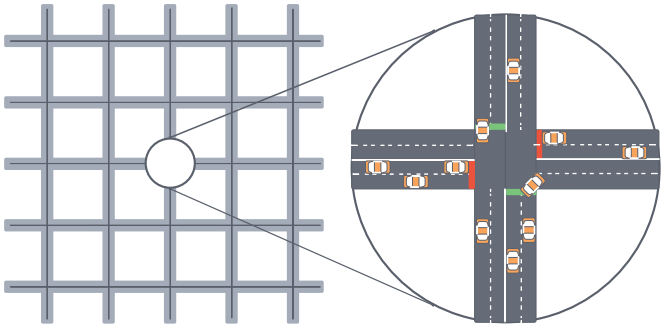


Fig. 2. A 5×5 traffic grid

To assess congestion in the traffic network, the state and reward is defined based on waiting time of vehicles and the queue length detected at all intersections. The local state $s_{t,i}$ is defined as $\{\text{wait}_t[l], \text{wave}_t[l]\}_{ji\in\mathcal{E}, l\in L_{ji}}$. Here l represents each incoming lane of intersection i, wait measures the cumulative delay of the first vehicle, and wave assesses the total number of approaching vehicles along each incoming lane within 50m to the intersection. The reward $r_{t,i}$ is formulated as $-\sum_{ji\in\mathcal{E}, l\in L_{ji}}(\text{queue}_{t+\Delta t}[l] + \alpha \cdot \text{wait}_{t+\Delta t})$, where queue is the measured queue length along each incoming lane, and α is a tradeoff coefficient. A longer queue length and increased waiting time for vehicles at intersections signify more severe

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traffic congestion, resulting in a smaller reward. The most optimal policy is for all intersections to collectively ensure a minimal length of queue and short waiting times throughout the entire traffic network, aiming to maintain the highest possible reward.

Traffic flows in real-world scenarios are characterized by their complexity in both spatial and temporal dimensions. To capture temporal dependencies in our scenario, we employ a Long Short-Term Memory layer as the final hidden layer of the DNN. This is the most straightforward method to involve feeding all historical states into an A2C agent. Since our proposed method focuses on aggregating agents' critic models, we train actor and critic DNN models separately. The experiment comprises 1 million training steps, each with a duration of 720 steps, resulting in approximately 1400 episodes. In our MDP, we set the discount factor γ to 0.99 and the tradeoff coefficient α to 0.75. The learning rates used in Algorithm 1 are configured as $\eta_\theta = 5 \times 10^{-4}$ for actor models and $\eta_\omega = 2.5 \times 10^{-4}$ for critic models.

B. Evaluation Metrics

In our experiment, we conducted a comparative analysis between A2FC, MA2C, and IA2C, which are all based on the A2C algorithm applied to MARL systems, particularly for ATSC. We evaluated these methods by measuring their training rewards throughout the training process. Additionally, we assessed the trained policies in terms of several key performance metrics, including average queue length, average intersection delay, and average vehicle speed within the traffic grid. Ideally, a well-trained policy should exhibit short and stable queue lengths, minimal intersection delays, and a consistent, higher vehicle speed.

C. Experimental Results

Fig. 3 plots the average training rewards of A2FC, MA2C, and IA2C against training time throughout the training process. These curves depict the upward trend of rewards that gradually converge as the agents accumulate experience and refine their local optimal policies. The data presented in this figure demonstrate that A2FC exhibits strong performance during the training process. While it may have a slightly slower convergence rate compared to MA2C, mainly because the centralized server aggregates agents' critic models only every 720 steps, A2FC ultimately converges to a higher reward, approximately -490. This convergent level is similar to MA2C's convergent reward of around -500 but significantly better than IA2C. Our method achieves a more stable convergence of rewards at the end of the training stage compared to MA2C's larger fluctuations.

Our method outperforms the others primarily due to its ability to provide a global view of the environment through the aggregation of agents' critic models. As previously explained, an agent's critic model encapsulates its comprehension of the reward function in the MARL system and its past environmental observations. The aggregation of these critic models imparts a comprehensive perspective of the environment and

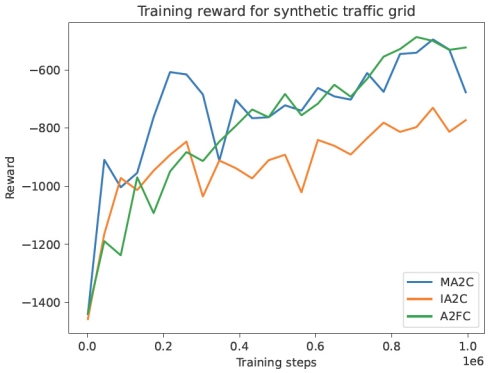


Fig. 3. Training reward

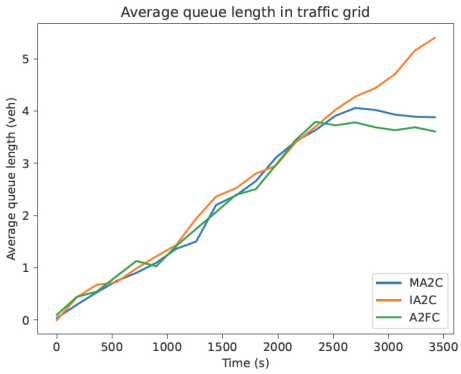


Fig. 4. Average queue length

the reward function to all agents. Consequently, agents can achieve convergence on a global scale.

In contrast, IA2C struggles to converge globally due to its limited communication among intersections, which is confined to a small region. MA2C enhances agents' environmental awareness by sharing more information about neighboring policies, but it still falls short of granting access to a complete global view. This limitation contributes to the fluctuations observed in MA2C's convergent rewards, as mentioned earlier. Additionally, MA2C necessitates extensive policy sharing among agents at each training step, resulting in high communication overhead compared to A2FC. Therefore, our proposed method excels in the training process, achieving a stable convergence to a higher reward while minimizing communication overhead.

Fig. 4 provides insight into the average queue length of all policies at each step in the experiment. Notably, IA2C exhibits a concerning trend with a continuously increasing average queue length, indicating its inability to effectively alleviate congestion in the system. Conversely, both A2FC and MA2C demonstrate more stable and sustainable policies. They achieve lower congestion levels and faster recovery. Fig. 5 illustrates the average intersection delay of the three methods. This metric follows a similar pattern to the queue length, with all methods experiencing an initial increase due to rising vehicle volume. However, in the later stages of testing, both A2FC and MA2C effectively decrease the average intersection delay, signifying their capacity to achieve traffic congestion recovery.

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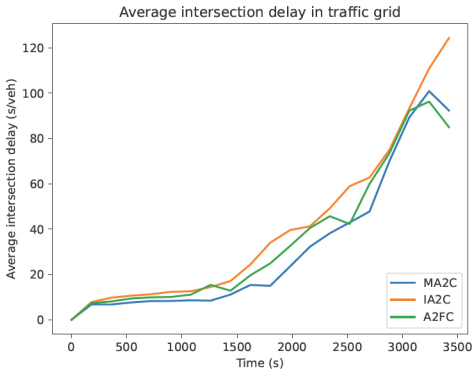


Fig. 5. Average intersection delay

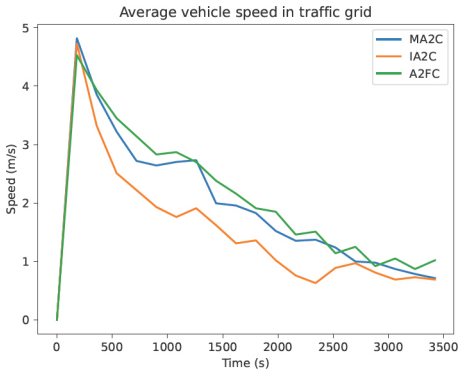


Fig. 6. Average vehicle speed

In contrast, IA2C struggles to mitigate congestion throughout the testing period. Our proposed method exhibits an advantage by achieving stability earlier and reducing the average intersection delay to a slightly lower level. This demonstrates the efficacy of A2FC in handling traffic congestion scenarios.

Metrics	Temporal Averages		
	MA2C	IA2C	**A2FC**
Avg. training reward	-738.3	-921.5	**-794.5**
Con. training reward	-499.4	-808.1	**-492.8**
Avg. queue length (veh)	2.4	2.7	**2.3**
Avg. intersection delay (s/veh)	31.7	40.4	**34.8**
Avg. vehicle speed (m/s)	1.9	1.5	**2.1**

TABLE I
ATSC PERFORMANCE OF ALL METRICS IN THE TRAFFIC GRID

Fig. 6 showcases the average vehicle speed achieved by the three trained policies. A2FC exhibits the highest average vehicle speed among the three policies. Despite a general decrease in average speed across all policies as vehicle volume increases, A2FC and MA2C demonstrate more stable performance and converge to a slightly higher average speed. This suggests that these policies are effective in reducing traffic congestion and improving vehicle flow. In contrast, IA2C lags behind, displaying the poorest performance of the three methods. This is primarily attributed to its slower convergence, resulting in a dramatic decrease in average vehicle speed and increased traffic congestion. Table I provides a summary of the experimental results, offering a comprehensive overview of the metrics for all three methods. It reinforces the superior performance of A2FC in mitigating congestion and enhancing traffic flow.

V. Conclusion

In summary, we have introduced a novel federated A2C framework known as A2FC, designed to address the challenges posed by agents with heterogeneous action spaces in MARL systems. A key feature of our approach is the utilization of a centralized server within the federated learning framework, which only aggregates agents' critic models, eliminating the need for communication among agents to share their policies. A2FC is well-suited for MARL systems where agents have diverse action spaces, offering substantial reductions in communication overhead as agents only exchange parameters related to critic models. Our method also prioritizes the privacy of agents during the training process. Our experiments conducted in an ATSC environment demonstrate the effectiveness of our method compared to other methods in MARL systems.

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